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Department of Mechanical Engineering

ME3 Literature Review Project

Ultrasonic guided wave propagation to characterise buried structures and detect defects in them

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Abstract

This literature review examined the application of ultrasonic guided waves in characterising structures and detecting defects. The fundamentals of bulk and guided waves were first established, recognising the long-range inspection capability of guided waves due to their confinement by waveguide boundaries. The four main attenuation mechanisms including intrinsic material absorption, scattering, geometrical spreading, and leaky wave losses were then explored. Leaky wave attenuation was identified to dominate in embedded waveguides, with soil porosity, compaction and saturation governing bulk and shear stiffness, thereby influencing impedance mismatch and attenuation magnitude. Attenuation was found to be a function of frequency, while finite element models tended to underestimate field values due to idealised assumptions. Guided wave interactions with defects showed that the refection coefficient increased with defect depth, axial and circumferential extent. Finally, practical considerations including transducer bandwidth, temperature variations and signal processing techniques were discussed.

Objectives

The objectives of this literature review are as follows:

1. Summarise and describe key properties of guided waves
2. Describe the effect of embedding waveguides in soil
3. Describe the scattering of guided waves at geometric features and defects
4. Discuss practical considerations that are relevant for field testing

The referencing style used in this paper is Vancouver referencing.

Generative AI tools were used in this report solely to improve phrasing, grammar, and conciseness. All technical content, analysis, and conclusions are the author’s own.

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1 Introduction

The safe operation of critical infrastructure such as pipelines, rail tracks, aircraft fuselage, and pressure vessels relies on the early detection of structural damage or defects. Flaws such as, cracks, notches, and corrosion can develop in these structures over time due to fatigue, environmental factors, and mechanical loading, necessitating timely intervention to prevent catastrophic failure. However, many of these defects are undetectable through visual inspections alone due to their inaccessible or sub-surface locations. Therefore, ultrasonic testing methods have been developed for the characterisation of structures and detection of defects. [1]

Traditionally, bulk waves have been employed in ultrasonic non-destructive testing, however, their use is limited by short inspection ranges and the requirement for direct surface access. As inspection applications evolve from non-destructive testing to structural health monitoring, a paradigm shift has occurred from bulk wave testing to guided wave techniques. Accordingly, this literature review examines the application of ultrasonic guided waves for the characterisation of buried structures and the detection of defects within them. (2)

This review first establishes the fundamentals of ultrasonic bulk and guided waves, followed by a comparative summary of their characteristics. The attenuation effects of embedding waveguides in soil are then examined, followed an analysis of guided wave behaviour when interacting with various defects, namely, through-thickness holes, notches, and corrosion pits. Finally, the review discusses practical consideration for field applications, including transducer performance and temperature effects on guided wave testing.

2 Properties of bulk and guided waves

Historically, ultrasonic wave testing mainly involved the use of bulk waves. However, as trends show a discernible shift from Non-Destructive Testing (NDT) to Structural Health Monitoring (SHM), the prevalence of Guided Wave Testing (GWT) is also on the rise [1].

The type of ultrasonic wave used influences the inspection methodology and instrumentation, resulting in varying resolution and inspection range.

2.1 Bulk Waves

As defined by Milosavljevic et al. [2], '*Bulk Waves exist in infinite homogeneous bodies and propagate indefinitely without being interrupted by boundaries or interfaces.*' (p.59) Bulk waves are typically generated at higher frequencies and therefore have shorter wavelengths than guided waves. Once generated by a point source, bulk wave energy spreads uniformly in all directions at a constant speed, forming a spherical propagation pattern. Consequently, the wave intensity decreases proportionally to the inverse square of the distance from the source. This geometric spreading combined with their higher frequency range, leads to rapid attenuation, limiting bulk waves to short-range applications. Additionally, the group and phase velocities of bulk waves are independent of frequency, making them nondispersive.

In isotropic media, bulk waves are categorised into two main types: longitudinal waves and transverse waves. Also known as pressure waves, longitudinal waves are polarised along their direction of propagation. Conversely, transverse waves, also known as shear waves, are polarised perpendicular to the direction of propagation. These transverse waves can be further classified into shear horizontal (SH) waves, where polarisation is out of the plane of propagation, and shear vertical (SV) waves, where the polarisation lies in the plane of propagation.

In anisotropic media, longitudinal and transverse waves are coupled, giving rise to quasi-longitudinal and quasi-shear waves whose polarisations are approximately parallel and perpendicular to the direction of propagation, respectively. As the shear stiffness of anisotropic media varies with direction, these quasi-shear waves are further classified into quasi-fast-shear waves, which propagate in the direction of higher shear stiffness, and quasi-slow-shear waves, which propagate in the direction of lower shear stiffness. Generally, quasi-longitudinal waves travel faster than both quasi-shear waves. [3]

2.2 Guided Waves

'*A guided wave is one whose wave propagation is guided by a structural form or boundary'* [4]. In plates, there are two types of guided waves: Lamb waves and Shear Horizontal waves, while in pipes, there are three types: longitudinal, flexural, and torsional [5]. In this section, each wave type will be characterised in terms of wave motion, dispersion curves, and mode shapes. Dispersion curves illustrate the phase or group velocity variation of each wave mode frequency component, while mode shapes represent the variations of displacement, stress, or other properties across the cross section of a structure.

2.2.1 Lamb Waves

Lamb waves are elastic waves that propagate within thin plates bounded by parallel, stress-free surfaces and are characterised by the coupling of longitudinal and shear motions [6]. Lamb waves are classified into symmetric (S) and anti-symmetric (A) modes. In symmetric modes, particle motion is a mirror image about the mid-plane, whereas in anti-symmetric modes the particles on the top and bottom surfaces move in opposite directions as shown in Figure 1 [7].

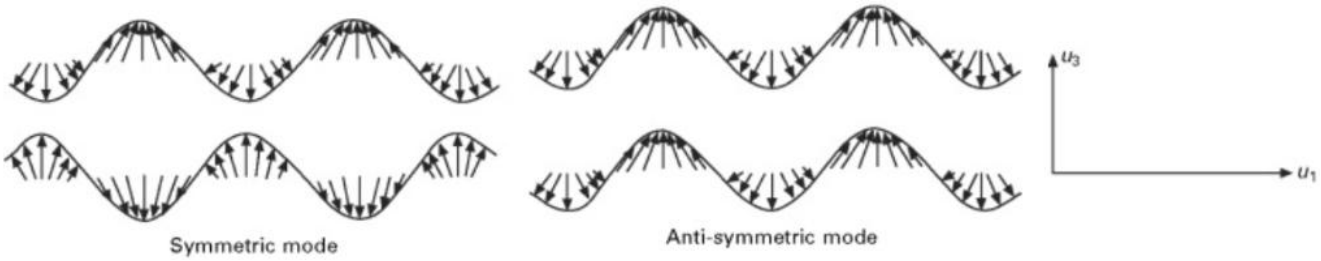


Figure 1 Symmetric and anti-symmetric Lamb modes [7]

Here u_1 denotes in-plane displacement along the direction of wave propagation and u_3 denotes out-of-plane displacement normal to the plate surface.

These symmetric and anti-symmetric modes are defined by the Rayleigh-Lamb Equations in an isotropic, homogeneous plate:

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{4k^2pq}{\left(q^2-k^2\right)^2} \text{ Symmetric Lamb wave modes} \quad (1)$$

$$\frac{\tan(qh)}{\tan(ph)} = -\frac{(k^2 - q^2)^2}{4k^2 qp} \quad \text{for anti-symmetric wave modes} \tag{2}$$

$$p^2 = \left(\frac{\omega}{c_L}\right)^2 - k^2 \text{ and } q^2 = \left(\frac{\omega}{c_T}\right)^2 - k^2 \tag{3}$$

Where *h* is half the plate thickness, *ω* is angular frequency, *k* is wavenumber, c_L longitudinal wave velocity, and c_T is shear wave velocity [8].

The dispersion curves for lamb waves in an aluminium plate are generated using the Dispersion software in Figure 2.

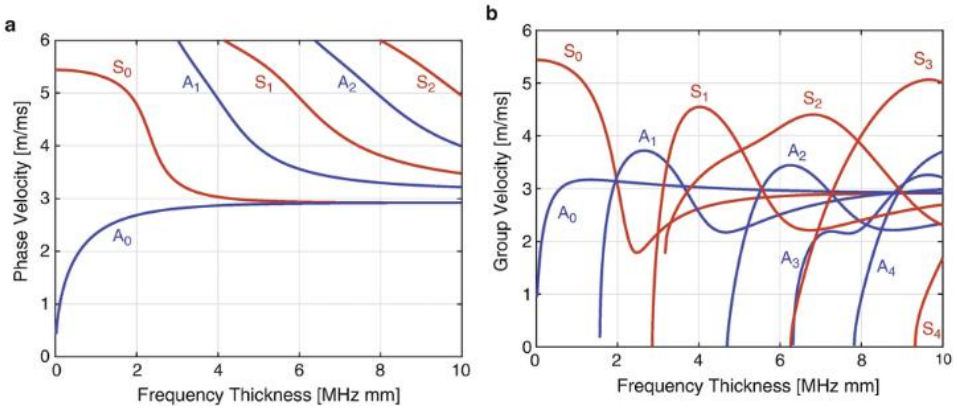


Figure 2 Dispersion curves for lamb waves in an aluminium plate (a) phase velocity (b) group velocity [9]

At their cutoff frequencies, each mode's phase velocity approaches infinity while group velocity approaches zero. With greater frequencies, the generation of additional Lamb wave modes results in greater overall dispersion. Dispersion refers to the phenomenon of different wave frequency components travelling at different velocities, leading to a change in the shape of the wave packet [9].

The S_0 and A_0 modes exhibit dominant in-plane and out-of-plane motion respectively, making these fundamental modes comparable to longitudinal and flexural waves at low frequencies as seen in Figure 3.

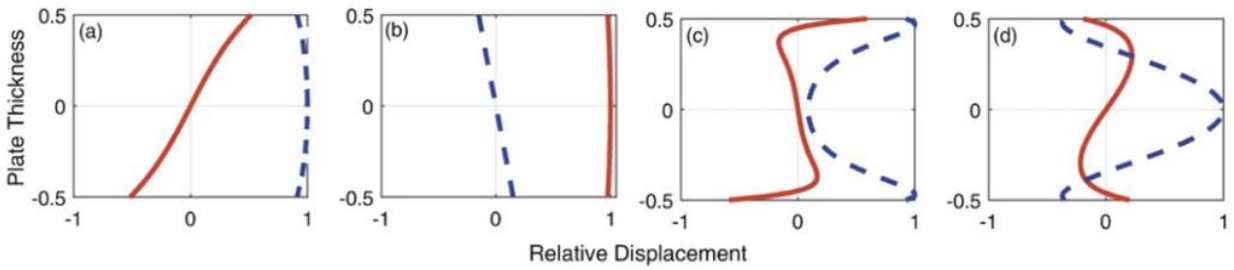


Figure 3 Lamb Wave mode shapes: (a) A0 mode at 0.5 MHz mm; (b) S0 mode at 0.5 MHz mm; (c) A0 mode at 10 MHz mm; (d) A1 mode at 10 MHz mm; solid: in-plane displacement; dashed: out-of-plane displacement [9]

2.2.2 Shear Horizontal Waves

Shear Horizontal (SH) waves are elastic waves that propagate within thin plates bounded by parallel, stress-free surfaces and are characterised by particle motion parallel to the plate surfaces and perpendicular to the direction of wave propagation. There are two types of SH modes: symmetric and anti-symmetric as illustrated by their mode shapes in Figure 4 [1] [10].

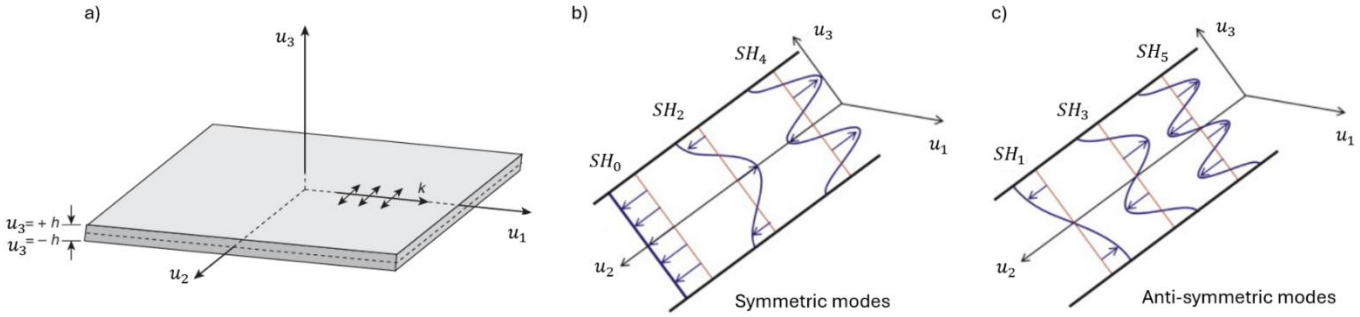


Figure 4 (a) SH wave propagation [1] (b) Symmetric SH wave mode shapes (c) Anti-symmetric SH wave mode shapes [10]

The phase, c_p , and group, c_g , velocities of SH waves in isotropic, homogeneous plates are defined by Equations 4 and 5:

$$c_p(fd) = 2c_T \left\{ \frac{fd}{\sqrt{4(fd)^2 - n^2 c_T^2}} \right\} \tag{4}$$

$$c_g(fd) = c_T \left\{ \sqrt{1 - \frac{(n/2)^2}{(fd/c_T)^2}} \right\} \tag{5}$$

Where f is frequency, d is the thickness of the plate and n is the mode number.

These equations yield the dispersion curves in Figure 5. Similar to Lamb wave modes at their respective cutoff frequencies, c_p approaches infinity and c_g approaches 0. As observed from the dispersion curves, the fundamental SH mode, SH_0, is nondispersive.

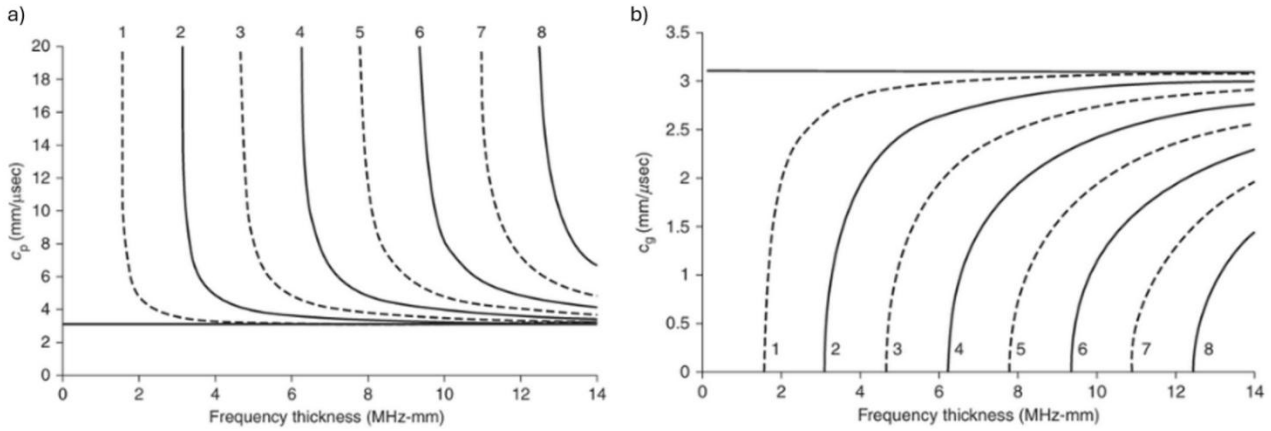


Figure 5 SH mode dispersion curves (a) SH mode phase velocity dispersion curves (b) SH mode group velocity dispersion curves; solid curves denote symmetric modes, and dashed curves denote antisymmetric curves [1]

2.2.3 Longitudinal, Torsional and Flexural Waves

The three types of guided waves in a hollow cylinder are longitudinal (L), torsional (T), and flexural (F) waves. Longitudinal and torsional modes are axisymmetric, while flexural modes are non-axisymmetric. Particle motion is dominant in the radial and axial directions for L modes, and in the azimuthal direction for T modes. F modes represent a combination of both L and T modes. These modes are visualised in Figure 6 [11].

The notation for these three modes is as follows: L(m, n), T(m, n), and F(m, n), where the integer m represents the circumferential order of a mode and the integer n represents the group order of a mode. For the axisymmetric modes L and T, m=0.

The dispersion curves for the L, T, and F modes are plotted in Figure 7[12], showing nondispersive behaviour in the T(0,1) mode across all frequencies and in the L(0,1) mode at lower frequencies.

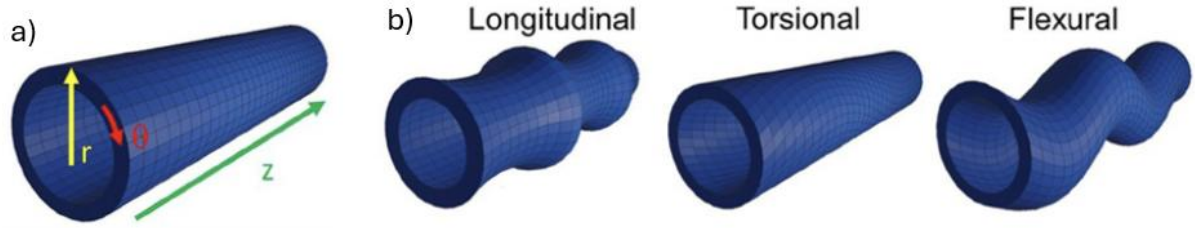


Figure 6 (a) Cylindrical coordinate system, showing the radial (r), circumferential (θ), and axial (z) directions (b) Visualisation of displacement behaviour of L, T, and F modes. [11]

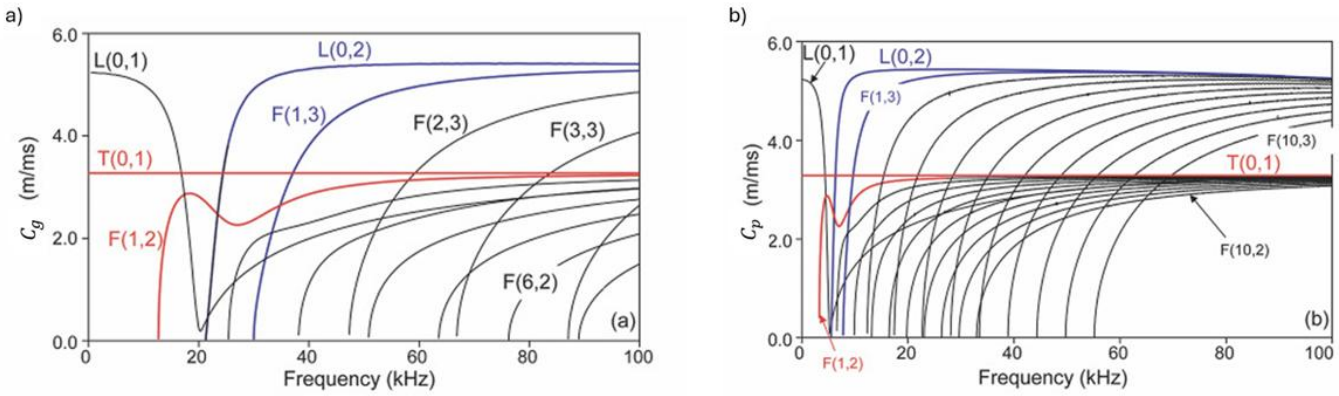


Figure 7 Dispersion curves for L, T, and F modes (a) Group velocity dispersion curve (b) Phase velocity dispersion curve [12]

2.3 A Comparison between Bulk and Guided Waves in Ultrasonic Testing

Ultrasonic testing (UT) is a NDT technique used to characterise structures and detect defects. Bulk and guided waves are the two primary waves employed in UT, each exhibiting distinct propagation behaviour and therefore suited to different inspection applications.

Bulk waves propagate freely within a structure without being affected by boundaries unless they encounter them. As bulk waves are generated at higher frequencies, they have shorter wavelengths that enable high spatial resolution. However, as bulk waves travel along narrow beam paths, inspection coverage is limited to the region directly beneath the transducer. Consequently, testing large or complex components can be time-consuming and impractical, as point-by-point scanning is required.

In contrast, guided waves are constrained by the waveguide's boundaries, allowing for long-range inspection from a single location. This makes guided wave techniques particularly useful for inaccessible or extensive structures. However, guided waves exhibit multimodal behaviour, resulting in dispersion and mode conversion that complicate signal processing.

Recent developments have sought to combine the strengths of both approaches. Advances in transducer design, signal processing, and numerical modelling have enhanced guided wave capability in ultrasonic NDT and SHM.

3 Guided Wave Attenuation

As ultrasonic guided waves propagate along structures, their amplitude decays with distance, a phenomenon known as attenuation. Understanding attenuation behaviour in buried structures is crucial for NDT and SHM, as energy losses reduce inspection range and can obscure defects.

Guided wave attenuation in embedded structures arises from four primary mechanisms: intrinsic material absorption, scattering, geometrical spreading, and energy leakage into the surrounding medium [13]. Intrinsic material losses arise from viscoelastic absorption of mechanical energy [14], while scattering occurs due to inhomogeneities within the waveguide [15][16]. Geometrical spreading refers to the reduction in wave amplitude that occurs as wave energy is distributed over an expanding propagation path [17]. In buried waveguides, however, attenuation is dominated by leaky wave losses, where a portion of the guided wave energy radiates into the surrounding medium.

This section focuses on leaky wave attenuation arising from the interaction between embedded waveguides and various soil conditions.

3.1 Leaky Wave Attenuation

Once a waveguide is embedded in another medium, the surface displacement of the propagating mode excites bulk waves in the surrounding medium [18]. Guided wave leakage occurs when the phase velocity of the guided mode exceeds the bulk velocity of the surrounding medium, allowing energy to propagate away from the waveguide and into the embedding material [19]. The propagation angle of these leaky waves is governed by Snell’s law, as formulated by Auld in 1990 [20]. On the other hand, the amplitude of the leaky wave is determined by the degree of acoustic impedance mismatch between the waveguide and the surrounding material [21].

3.2 Influence of Soil Conditions on Leaky Wave Attenuation

The attenuation of guided wave propagation under different soil conditions has been examined using both experimental and numerical methods. Table 1 summarises the attenuation data from three key studies.

Leinov et al. (2015) [22] conducted laboratory experiments on buried steel pipes, using Disperse [23] to validate results against known soil acoustic properties. In contrast, Su et al. (2021) [24] employed a finite element model and compared their predictions with attenuation trends reported by Leinov et al. (2015). However, since this approach assumed idealised, fully saturated soils, its applicability to real field conditions was limited. Long et al. (2017) [25] performed field tests on iron water pipes in natural soils, capturing realistic boundary conditions but lacking control over soil composition.

Collectively, these studies highlight how soil properties such as saturation, porosity, and compaction influence soil impedance, which in turn governs the impedance contrast between the pipe and embedding medium, thereby determining the magnitude of guided-wave attenuation.

Table 1 Summary of reported attenuation coefficients in literature for pipes buried under various soil conditions.

Soil Condition	Density (kg/m^3)	Attenuation (dB/m)	Mode	Frequency (kHz)	Pipe	Reference
Loose	1455	2.0-3.7 1.1-1.5	T(0,1) L(0,2)	11-34	Carbon steel pipe, r_0=110.6mm, t = 9mm	[26]
Compacted	1620	3.4-4.9 2.0-2.5	T(0,1) L(0,2)			
Mechanically compacted	1660	3.2-4.5 2.4-3.2	T(0,1) L(0,2)			
Saturated loose	1895	1.7-2.3 0.98-1.4	T(0,1) L(0,2)			
Saturated Compact	1895	2.4-3.2 1.5-2.0	T(0,1) L(0,2)			
Drained	1724	4.0-5.5 2.0-2.7	T(0,1) L(0,2)			
Infinite porous medium	β = 0.1 β = 0.2 β = 0.3	103.2 ≈ 130.0 ≈ 225.0	L(0,1)	45	Steel pipe, water-filled, r_0 = 16 mm, t = 2 mm	[24]
Finite porous medium of depth 0.05 m	β = 0.1 β = 0.2 β = 0.3	1.8 ≈ 5.0 ≈ 9.0				
Greenwich clay	1950	≈ 17.0-38.0	L(0,1)	50-250	Ductile iron pipe, water-filled, r_0 = 83.2 mm, t = 7 mm	[25]
Guildford clay	1900	≈ 15.0-28.0				
Alnwick soil	2100	≈ 13.0-19.0				

3.2.1 Influence of Soil Structure

The soil structure refers to the arrangement of soil particles and the distribution of voids between them. Porosity, β, as defined by Su et al. (2021) is the ratio of the volume of voids to the total volume of the soil.

Santamarina (2001) [27] demonstrated that longitudinal wave velocity depends on both bulk and shear stiffness, whereas shear velocity only depends on the latter. Shear stiffness, governed by inter-particle forces and material type, is typically much lower than bulk stiffness. In fully saturated soils, bulk stiffness dominates and varies with porosity. The presence of air greatly reduces bulk stiffness, causing wave velocity to be controlled primarily by shear stiffness.

Santamarina's findings are supported by Leinov et al. (2015), who experimentally observed that increasing soil compaction, which increases both bulk and shear stiffness, led to higher guided wave attenuation. Since acoustic impedance is proportional to the product of density and wave velocity, this trend supports that a smaller impedance mismatch leads to stronger coupling between the pipe and surrounding soil, hence greater guided wave attenuation.

Conversely, Su et al. (2021) contradicted Santamarina's findings. Using a Biot-based model of fully saturated porous soils [24], they found that attenuation increased with porosity. As greater porosity reduces bulk stiffness, these results oppose Saantamarina's assertion that bulk stiffness dominates in saturated soils. However, Su et al. (2021) justified their findings by attributing the rise in attenuation to increased viscous losses across a greater soil-fluid interface.

3.2.2 Influence of Soil Saturation

Leinov et al. (2015) found that the attenuation for drained sand was approximately equal to that of mechanically compacted sand, while saturated sand exhibited higher attenuation than loose sand but lower than drained sand. This trend supports Santamarina's (2001) assertion that the presence of air greatly reduces bulk stiffness and that bulk stiffness dominates in saturated soils.

Similarly, Long et al. (2002) described Greenwich soil as having a “sticky consistency”, Guildford clay as “torn apart easily”, and Alnwick soil as “drier than the other two”. Combined with their reported density values, this suggests an increasing moisture content across these soils, corresponding to higher attenuation values as air content decreases. While this trend broadly supports Santamarina’s claim, the qualitative nature of Long et al.’s soil descriptions and uncontrolled in-situ conditions limit the reliability of this conclusion, highlighting the need for quantitative validation.

Both studies affirm the dependence of attenuation on soil saturation, as changes in bulk stiffness alter the acoustic impedance contrast between the pipe and surrounding soil.

3.3 Influence of Wave Parameters

All three key studies have shown that attenuation increases with frequency. Leinov et al. (2015) additionally observed that the T(0,1) mode consistently exhibited greater attenuation than the L(0,1) across all frequencies and soil conditions, attributed to its larger radial displacement and stronger coupling with the surrounding medium. Su et al. (2021) confirmed this frequency dependence, showing near-linear increase in attenuation with frequency. However, their Biot-based model assumes perfect soil-pipe coupling and neglects surface roughness, which may lead to an underestimation of attenuation in field conditions.

4 Guided Wave Interactions with Geometric Features and Defects

Upon encountering geometric features or defects, guided waves are scattered and propagate in directions different from their original path [28]. The incident wave may be reflected or transmitted [29], leading to wave superposition and possible mode conversion. The resulting scattered fields are then analysed to characterise defects and other structural features [30]. Parameters such as the defect length relative to the wavelength [31], the proportion of wall loss in the cross-section of the component [32], and the defect orientation [33] can be inferred from the reflection amplitude. Similarly, the arrival time of the reflected signal indicates the distance of the defect from the

transducer, while differences in the arrival times of reflected components can help distinguish between different wave modes [34].

In pipes, the symmetry of the reflected guided wave modes is indicative of the symmetry of the defect [35]. Axially symmetric defects such as flanges, square ends, or uniform welds (with end caps) predominantly generate axisymmetric mode reflections. In contrast, non-axisymmetric defects such as corrosion pits result in some non-axisymmetric modes being generated [21]. Hence, this section will examine the scattering behaviour holes, notches, and corrosion pits.

4.1 Through-Thickness Holes

In Løvstad's (2012) thesis [34], the scattering field and mode conversion of S_0 and SH_0 modes incident on a 60 mm-diameter through-thickness hole was modelled using finite element analysis. The model included absorption regions to eliminate edge reflections and was validated against the method outlined by Diligent et al. (2002) [36]. As shown in both Figure 8 (a) and (b), the incident plane waves propagate in the z-direction and interact with the hole to produce distinct reflected and transmitted wavefronts. For SH_0 wave incidence in Figure 8 (a), conversion to S_0 occurs, whereas for S_0 incidence in Figure 8 (b), conversion to SH_0 is observed. In both cases, the S_0 and SH_0 mode reflections are clearly distinguishable as S_0 exhibits a greater group velocity than SH_0.

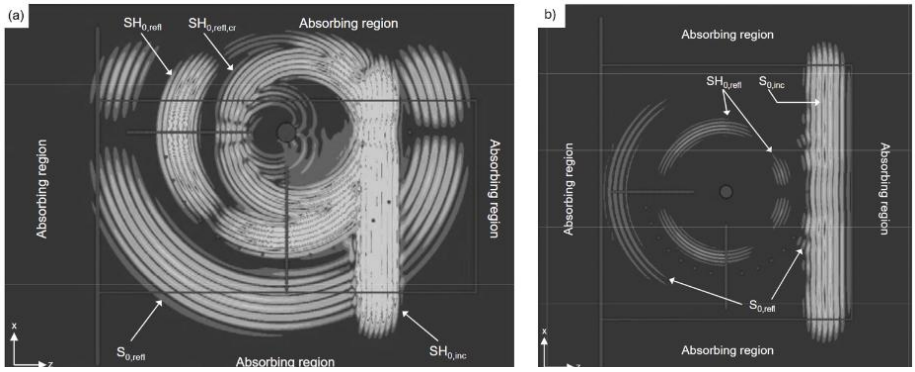


Figure 8 Wave propagation and scattering with plane (a) SH0 wave incidence and (b) S0 wave incidence [34]

The finite element (FE) results revealed that reflection amplitude increased with hole diameter, with both scattered modes exhibiting strong angular dependence. The scattered SH_0 mode consistently had a greater amplitude than the S_0 mode, partly due

to its shorter wavelength relative to the defect size. Both modes were symmetric about the propagation axis, with SH_0 and S_0 dominating in the forward-scattering and backward-scattering directions, respectively [34].

Løvstad's numerical findings generally agreed with the analytical predictions by Cegla et al. (2008) [37], confirming the dependence of reflection amplitude on defect size relative to wavelength. However, for larger holes, the FE model diverged from the analytical solutions due to the emergence of additional circumferential modes not captured in the concatenated solutions. This highlights a key limitation of analytical approaches for three-dimensional pipe geometries, underscoring the importance of accounting for multi-mode scattering effects in guided wave inspections.

4.2 Notches

The reflection coefficient (RC) due to notch defects in pipes is dependent on its depth and circumferential extent. Unlike through-thickness notches, the axial extent of part-depth notches also significantly influences the RC. These relationships were explored by Demma et al. (2004) [38] using axi-symmetric and three-dimensional FE models, experimentally validated in earlier work by Demma et al. (2003) [39]. In these studies, the RC was defined as the ratio of the reflected to the incident amplitude.

The interference between reflections from the front and back faces of a rectangular notch, Figure 9(a), results in the periodic relationship between the RC and relative axial extent as shown in Figure 9 (c). The maximum RC, R_{max}, is produced by constructive interference when the reflections are in-phase, whereas the minimum RC, R_{min}, is a result of destructive interference when the reflections are out-of-phase.

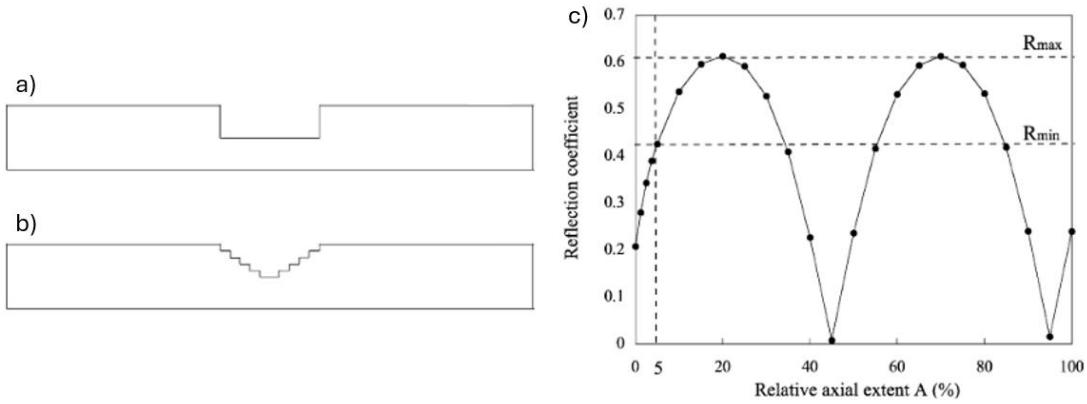


Figure 9 (a) Schematic of rectangular notch (b) schematic of stepped notch (c)Reflection coefficient for axisymmetric notches of varying axial extent. Results are for T(0,1) incident on a 24 in. pipe at 35 kHz and b = 0.5t (50% depth notch). [38]

While the axial extent influences the interference pattern, the circumferential extent determines the symmetry of the reflected modes. As shown in Figure 10 (a) and (b), when an axisymmetric mode is incident upon an axisymmetric defect, only axisymmetric (A) reflected modes are produced. Conversely, when the defect is non-axisymmetric, anti-symmetric flexural (F) reflected modes are produces alongside the A modes.

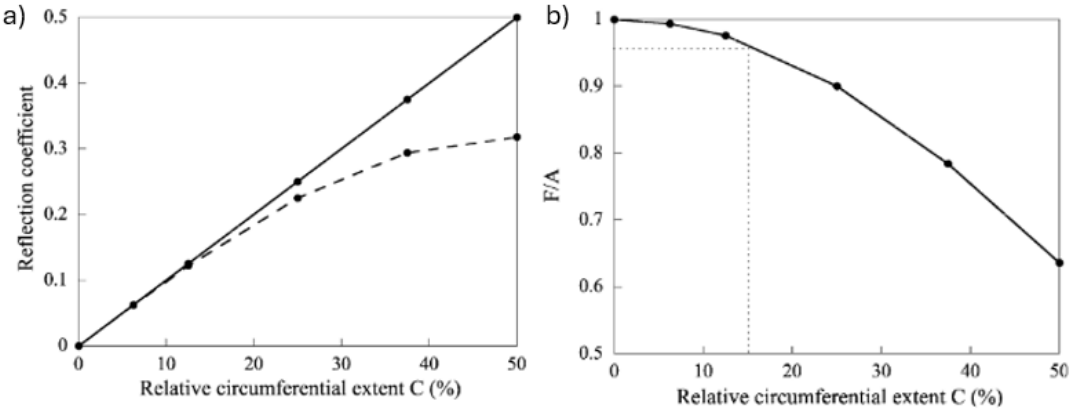


Figure 10 (a) Reflection coefficient for axisymmetric (solid line) and flexural mode of the first order (dashed line) as a function of the circumferential extent of a through thickness defect. (b) Ratio between F and A mode reflection at varying circumferential extent. [38]

Axi-symmetric FE results demonstrated that RC increased with defect depth. For axisymmetric defects, Demma et al. (2004) displayed the RC as functions of defect depth and axial extent in Figure 11 (a) and (b). For anti-symmetric defects, the RC was mapped as a function of depth and circumferential extent as shown in Figure 11 (c) and (d). The axial extent in this case was inferred from the F/A ratio in Figure 10 (b). Overall,

the study confirmed that RC increases with increasing notch depth, axial extent, and circumferential extent.

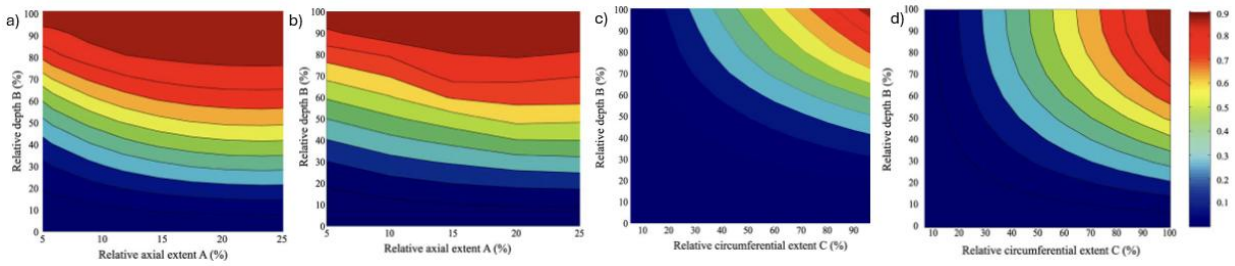


Figure 11 Map of reflection coefficient from axisymmetric defects with varying depth and axial extent. (a) T(0,1) (b) L(0,2) (c) Minimum reflection from defects with varying depth and circumferential extent for T(0,1) (d) Maximum reflection from defects with varying depth and circumferential extent for T(0,1) [38]

Fromme (2010, 2017) [31] [33] extended these relationships to plate structures and found comparable trends to Demma et al. (2004). Using A_0 mode excitation, Fromme observed that the reflection amplitude increased with notch depth and length relative to wavelength, with strong angular dependence of scattering amplitude. Scattering was minimal along the axial direction of the notch and greatest in the incident wave plane. However, Fromme's study also highlighted that analytical models tend to underestimate RC for shallow or irregular notches, as the simplified models assume sharp, through-thickness discontinuities.

Overall, the findings by Fromme (2010, 2017) and Demma et al. (2004) affirm that defect geometry and orientation govern the scattering intensity. The agreement between experimental results and FE predictions reinforces the reliability of numerical modelling for notch characterisation. However, discrepancies for smaller and irregular defects highlight the limitations of analytical approaches.

4.3 Corrosion and Pits

To better understand the scattering effect of corrosion defects, representative geometries are often used for analysis. Demma et al. (2004) modelled corrosion patches as stepped notches, Figure 9(b), whereas Løvstad (2012) examined pit clusters. As shown in Figure 14 (a), stepped notches produced RC trends similar to rectangular notches, with approximately 20% lower maximum RC.

Løvstad (2012) identified a linear relationship between RC and the total volume fraction of material loss as shown in Figure 12 (b). However, smaller, deeper, or axially spaced pits exhibited destructive interference between reflected wavefronts, leading to smaller RC values. When pit clusters are superimposed on cosine-tapered corrosion profiles, the maximum RC occurred at a quarter wavelength, consistent with findings by Demma et al. (2004) [38].

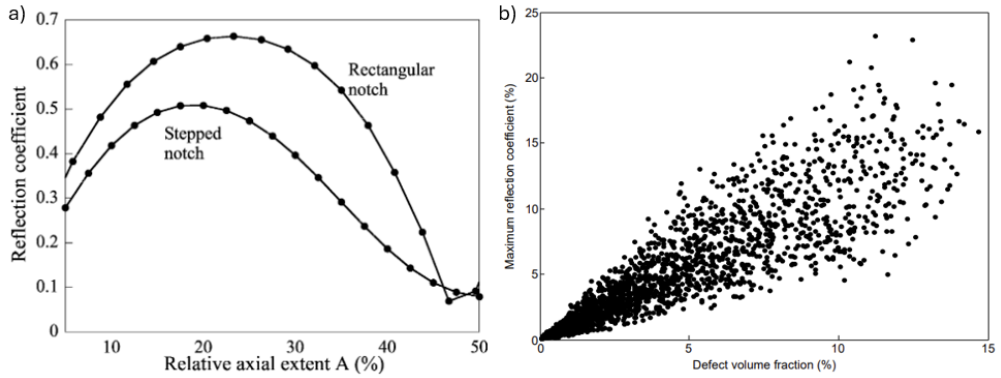


Figure 12 (a) RC against axial extent for T(0,1) [38] (b) Maximum RC against defect volume fraction [34]

Collectively, both studies show agreement that RC increases with total volume material loss. However, each modelling approach has its inherent limitations. Demma et al.'s (2004) stepped-notch model may overestimate RC values due to its well-defined edges, whereas Løvstad's (2012) pit-cluster model more realistically captures the distributed nature of corrosion but is constrained by idealised pit shapes. Hence, these discrepancies indicate that hybrid modelling strategies that integrate numerical methods with experimental validation are required to realistically capture corrosion.

5 Practical Considerations for Field Applications

5.1 Guided Wave Transducers

Ideally, guided wave transducers are designed to excite a single mode to simplify defect detection and signal processing. This ideal scenario assumes an infinite and continuous plate excitation that produces a specific phase velocity and hence a pure guided mode (1). However, in practice, transducers are of finite size and bandwidth, resulting in distributed loading over a limited region across multiple wavenumbers, which generates various modes simultaneously. The generation of specific modes becomes challenging

when neighbouring modes with similar phase velocities [35], producing coherent noise that lowers the system's signal to noise ratio (SNR) and overall sensitivity.

To address this, practical methods have been employed to mitigate this problem. Transducer array configurations have been used to enhance the amplitude of desired modes through controlling the spacing, phasing, and pitch of individual elements. This setup exploits the principle of interference to narrow the bandwidth of the desired mode. However, this method still retains some unwanted modes.

Additional techniques include frequency tuning and digital signal processing. Signal processing methods such as 'focusing' by Rose et al. (2010) [40] or mode isolation in the frequency domain by Li et al. (2023) [41] have proven to improve mode selectivity. Another effective approach is the careful frequency range selection, in a region of the dispersion curve where only one or few modes exist.

5.2 Temperature Effects

Temperature variations impact the performance of guided wave testing and structural health monitoring [42] This phenomenon is mainly attributed to the changes in material properties, including elastic modulus and density of both the waveguide and the transducer [43]. Variations in waveguide material properties alter the phase and group velocities, which in turn affect the time-of-flight and phase of the received signals [44].

A study by Dhutti et al. (2018) [45] found that thermal expansion between the transducer and pipe surface reduced coupling efficiency, leading to reduced signal transmission. Expanding on this, Ren et al. (2024) [46] investigated the influence of elevated temperature changes on different transducer configurations and established that embedded PZTs were less affected by temperature changes than their surface mounted counterpart [46].

In field applications, these effects are mitigated through the incorporation of temperature-compensation algorithms and baseline subtraction methods that account for wave velocity changes. Additionally, thermally stable piezoelectric materials are also chosen to minimise coupling inefficiencies between the transducer and waveguide [47].

6 Conclusion

This literature review examined the application of ultrasonic guided wave propagation for characterising structures and detecting defects. The primary guided waves in plates are Lamb and Shear Horizontal waves, while in pipes they are Longitudinal, Torsional and Flexural modes. Guided waves have a key advantage to bulk wave due to their long inspection range as a result of being constrained by waveguide boundaries. However, due to their multi modal behaviour, signal processing becomes more complicated.

Guided wave attenuation was shown to arise from four primary mechanisms: intrinsic material absorption, scattering, geometrical spreading, and energy leakage into the surrounding medium. Leaky wave attenuation proved to be dominant in embedded waveguides. When buried in soil, attenuation was found to strongly depend on soil properties and wave parameters. Soil properties such as porosity, compaction, and saturation govern bulk and shear stiffness, which in turn controls the impedance mismatch, hence attenuation. Experimental and numerical studies confirmed that attenuation increased with frequency. However, through idealised assumptions, finite element models tended to underestimate field attenuation.

Investigations into the interaction of guided waves with defects revealed that the reflection coefficient increases with defect depth, axial length, and circumferential extent. Finite element analysis reliably captured scattering the trends observed experimentally, though numerical predictions tended to underestimate RC for shallow or irregular defects. Corrosion defects saw an increase in RC with total volume of material loss.

Finally, several practical challenges in field applications were identified, including unwanted mode generation due to transducer bandwidth, temperature-induced coupling inefficiencies, and the need for additional signal processing. Developments such as array transducers and temperature compensation algorithms show promise in mitigating these issues and improving guided wave reliability in practical applications.

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